Periodontal Breakdown Cascade

Integrated Biomedical and Clinical Sciences 2 OMFS 513

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Mission: To advance the science and art of health through education, service, scholarship and social accountability

Learning Objectives & Competency Statements

Learning Outcomes

1. Understand the periodontal breakdown cascade
2. Understand the pathogenesis of periodontal disease

Competency statements

Define and integrate the knowledge of periodontal breakdown cascade and its impact on periodontium’s health and disease

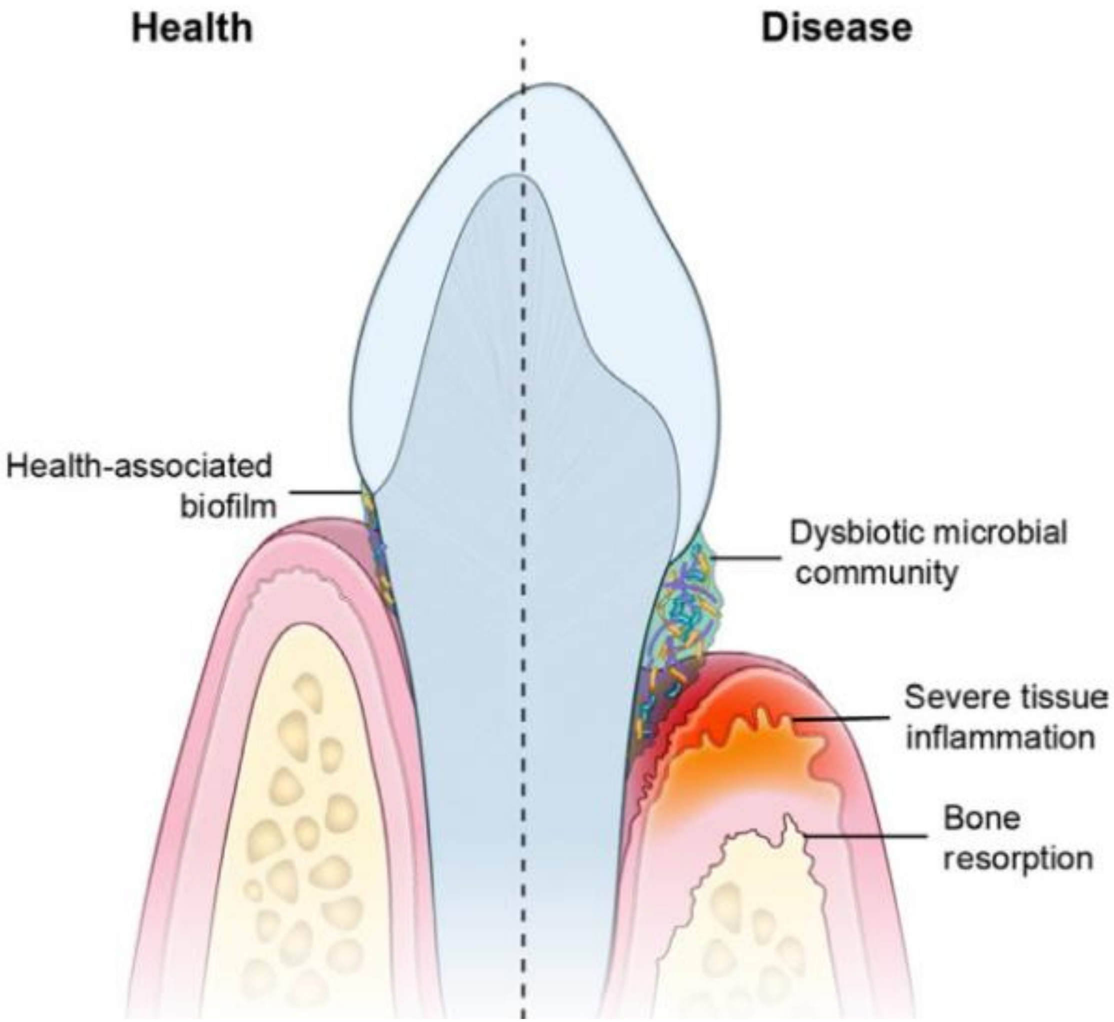
Pathogenesis

-Pathogenesis:

Origin and development of disease; the step-by-step process by which etiologic factors cause structural and functional changes in periodontium

-Inflammatory Responses in Periodontium:

- Tissue damage signals released during inflammation
- **Two sources:**
    - Subgingival microflora (microbial virulence factors)
    - Host immune-inflammatory response
- **Majority** of tissue breakdown is due to **host response**
- Bacteria initiate and sustain inflammation, but cause **limited direct** damage



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Pathogenesis

-Microbial Virulence Factors: subgingival biofilm initiates/perpetuates inflammatory responses in gingival/periodontal tissues

-Subgingival bacteria contribute directly to tissue damage by releasing noxious substances, their primary importance in periodontal pathogenesis is **activating immune-inflammatory responses** resulting in tissue damage

-**Lipopolysaccharide:** large molecules made of lipid and polysaccharide in outer membrane of gram-negative bacteria (important in structural integrity of bacterial cells), acting as endotoxins, eliciting strong immune responses

Pathogenesis

-Immune system recognizes LPS via **toll-like receptors** (TLRs), a family of cell surface molecules, important in innate immune responses

-TLRs recognize **microbe-associated molecular patterns** (MAMPs): molecular structures on pathogens. This interaction triggers intracellular events, resulting in increased **inflammatory mediators (most importantly cytokines)** and immune cells differentiation for effective immune responses against pathogens

-A component of **gram-positive cell walls**, **lipoteichoic acid** (LTA), also stimulates immune responses (**less potently than LPS**)

-Both LPS and LTA are released from biofilm bacteria and stimulate inflammatory responses, increasing vasodilation and vascular permeability, recruiting inflammatory cells by chemotaxis, and releasing proinflammatory mediators by leukocytes

-LPS is important in initiating and sustaining inflammatory responses in gingival/periodontal tissues

Bacterial Enzymes and Noxious Products

- Bacteria produce metabolic waste products contributing directly to tissue damage affecting host cells
- **Fatty acids** may aid P. gingivalis infection through tissue destruction and create a nutrient supply for organism by increasing bleeding into periodontal pocket
- Fatty acids also influence cytokine secretion by immune cells and potentiate inflammatory responses after exposure to proinflammatory stimuli such as LPS, IL-1β, TNF-α
- Plaque bacteria produce **proteases** (breaking down periodontium structural proteins: collagen, elastin, fibronectin)

Bacterial Enzymes and Noxious Products

- Proteases digest proteins and provide peptides for bacterial nutrition

- Bacterial proteases disrupt host responses, compromise tissue integrity, and facilitate microbial tissue invasion

- **Microbial Invasion**: histologically bacteria (cocci, filaments, rods) are identified in intercellular spaces of epithelium

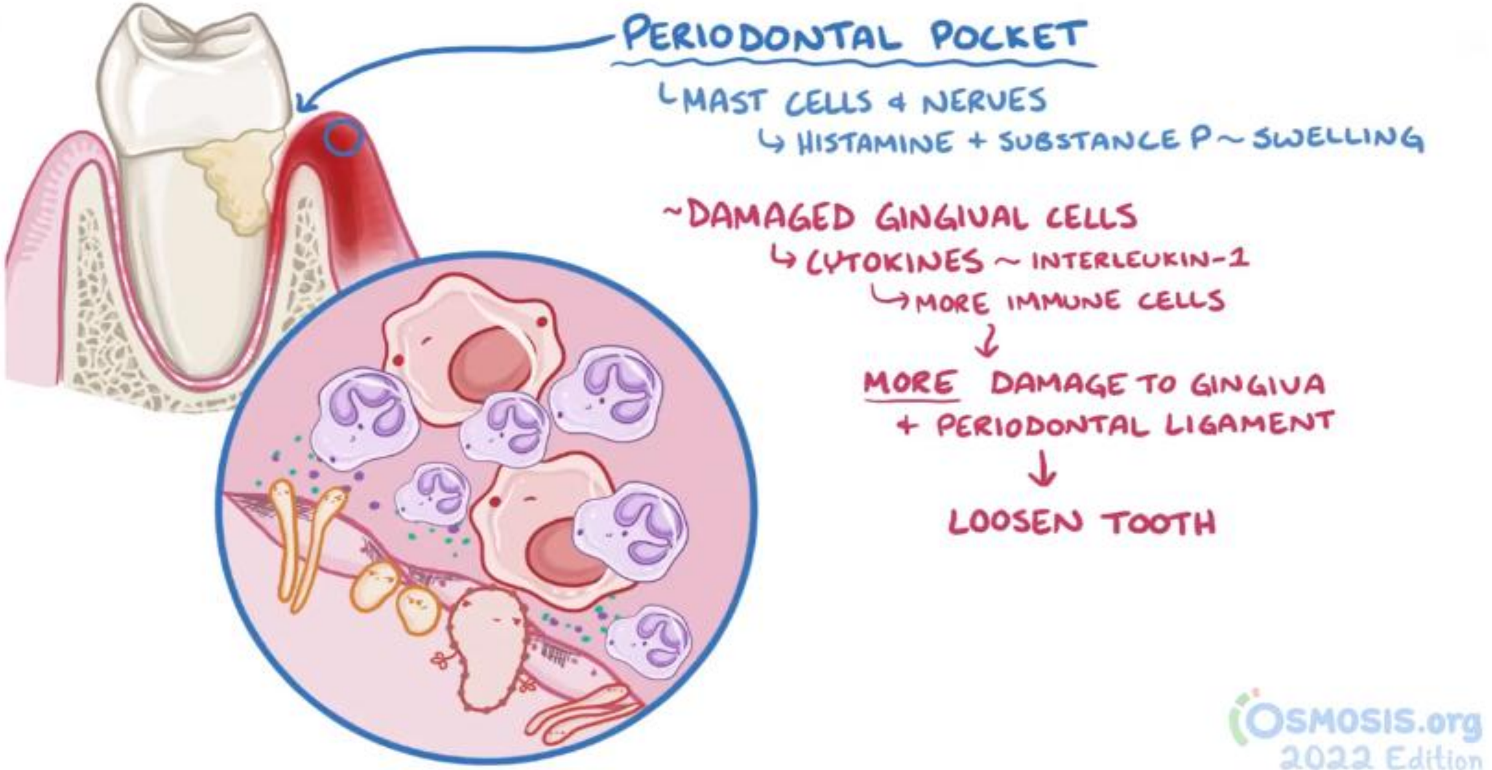
- Does bacterial presence in tissues **justify antibiotics use?** (in treating periodontitis)

- Bacteria in tissues is a "reservoir for reinfection". However, until clinical relevance of bacteria being present in tissues is better defined, it is inappropriate to make clinical treatment decisions (e.g., to use **adjunctive systemic antibiotics**)

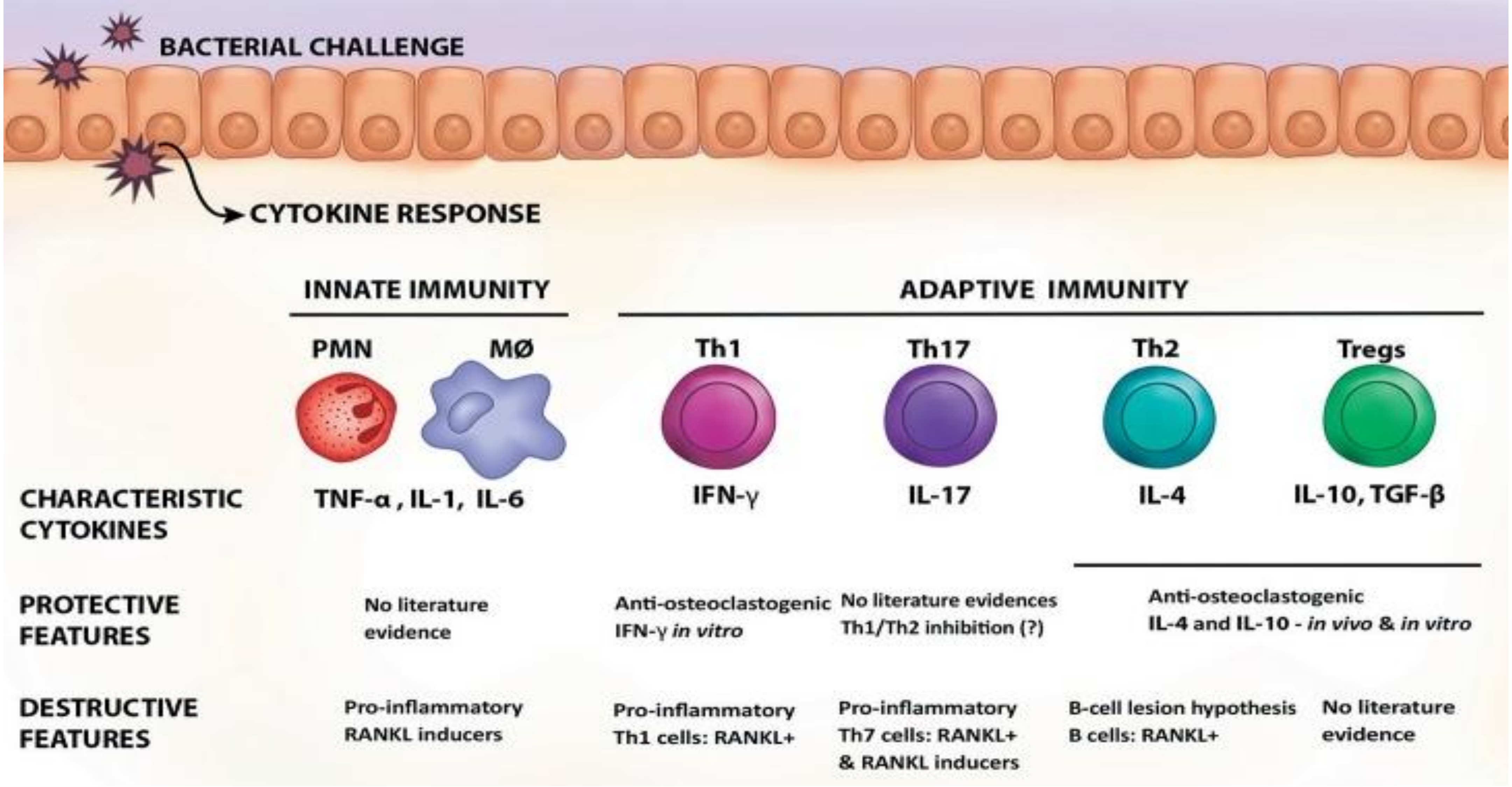
Host-Derived Inflammatory Mediators

-Host-Derived Inflammatory Mediators: inflammatory and **immune processes** that develop in periodontal tissues in response to subgingival biofilm are **protective by intent** but **result in considerable tissue damage**. **Host response is mainly responsible for tissue damage, leading to clinical signs/symptoms of periodontal disease**. Majority of tissue damage in periodontitis derives from **excessive** and **dysregulated** production of inflammatory mediators and destructive enzymes in response to presence of subgingival plaque bacteria.

-Key types of mediators orchestrating host responses: cytokines, prostanoids, matrix metalloproteinases (MMPs)



Cytokines & Periodontal Disease



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-Cytokines: play a fundamental role in inflammation

-key inflammatory mediators in periodontal disease

-Are **soluble proteins acting as messengers to transmit signals from one cell to another** and **bind to specific receptors on target cells** and **initiate intracellular signaling cascades resulting in phenotypic changes in the cell**

-**Produced by:** inflammatory cells: neutrophils, macrophages, and lymphocytes/ periodontium resident cells: fibroblasts and epithelial cells

-**Signal, broadcast, amplify immune responses**, are **important in regulating immune-inflammatory responses** and in combating infections

-Have biologic effects **causing tissue damage in chronic inflammation**

-Prolonged/excessive production of cytokines and other inflammatory mediators in periodontium causes tissue damage characterizing clinical signs of disease. e.g, cytokines mediate connective tissue and alveolar bone destruction through fibroblasts and osteoclasts induction to produce proteolytic enzymes (i.e., MMPs), breaking down structural components of connective tissues

-Proinflammatory cytokines **IL-1β** and **TNF-α** play a key role in initiation, regulation, and perpetuation of innate immune responses in periodontium, resulting in **vascular changes** and **migration of cells** such as **neutrophils** into periodontium as part of a normal immune response to subgingival bacteria

-Once the immune and inflammatory processes are initiated and the complex cytokine network is established, various inflammatory molecules that play a direct role in periodontium degradation are produced in response to **MAMPs** and/or **host-derived cytokines**

Prostaglandins

- A group of **lipid** compounds derived from arachidonic acid, a polyunsaturated fatty acid found in cell plasma membrane

- **Important inflammatory mediators, particularly (PGE2)**, results in **vasodilation** and induces **cytokine production**

- PGE2 is **produced by** a variety of cells and mostly in periodontium by **macrophages** and **fibroblasts**

- PGE2 results in **induction** of **MMPs** and **osteoclastic bone resorption** and has a major role in tissue damage that **characterizes periodontitis**

Matrix Metalloproteinases

- **MMPs:** a family of **proteolytic enzymes** that **degrade extracellular matrix** molecules: **collagen, gelatin, elastin**
- **Proteases: zinc/calcium-dependent** with an essential role in extracellular matrix **(ECM) turnover** and **degradation**. MMPs degrade all ECM components
- **Produced** by a variety of cells: **neutrophils, macrophages, fibroblasts, epithelial cells, osteoblasts, osteoclasts**
- Are **inhibited** by **tetracycline**, led to development of antimicrobial formulation of **doxycycline** as an adjunctive drug treatment for periodontitis using the **anti-MMP properties** of this molecule

Inflammatory Mediators in Periodontal Disease

- **Interleukin-1 Family Cytokines**
- **Tumor Necrosis Factor Alpha**
- **Interleukin-6**
- **Prostaglandin E2**
- **Matrix Metalloproteinases**

- **Neutrophils:** key infiltrating cells in periodontitis, large numbers in inflamed periodontal tissues, responding rapidly/aggressively to external stimuli (bacterial LPS), releasing large quantities of destructive enzymes rapidly
- Predominant MMPs in periodontitis, **MMP-8** and **MMP-9**, **secreted** by **neutrophils**, very effective at **degrading type 1 collagen** (the most abundant collagen type in periodontal ligament)
- **Prolonged/excessive release of MMPs** in periodontium leads to **significant breakdown** of **structural components** of **connective tissues**, contributing to clinical signs of disease
- MMPs are important in **alveolar bone destruction** and **expressed** by **osteoclasts**

Linking Pathogenesis to Clinical Signs of Disease

- **Advanced periodontal disease** are characterized by **tooth mobility** from **loss of attachment between tooth** and its **supporting tissues** following **breakdown of periodontal ligament inserting fibers** and **alveolar bone resorption**

- **Epithelial cells** in **sulcular epithelium** are in **constant contact** with bacterial products and respond by **secreting chemokines** to attract **neutrophils**

- If bacterial challenge persists, **cellular** and **fluid infiltration** continues to develop, and neutrophils and other inflammatory cells occupy a significant volume of inflamed gingival tissues

- **Neutrophils:** protective leukocytes that phagocytose/kill bacteria, key components of **innate** immune system, play a role in maintaining periodontal health despite constant challenge by plaque biofilm

Linking Pathogenesis to Clinical Signs of Disease

- **Deficiencies** in **neutrophil** functioning result in **increased susceptibility** to infections
- They **release large quantities of destructive enzymes, such as MMPs** resulting in breakdown of structural components of periodontium
- Neutrophils release **potent lysosomal enzymes, cytokines,** and **reactive oxygen species** extracellularly, causing further tissue damage (continued tissue damage and collagen depletion in periodontal tissues and the epithelium continues to proliferate apically, deepening the pocket, which is colonized by subgingival bacteria)
- **Chronic inflammation cycle:** presence of **subgingival bacteria** drives **inflammatory responses** in periodontal tissues, characterized by **leukocytes infiltration**, **inflammatory mediators** and **destructive enzymes release**, **connective tissue breakdown**, and **breakdown/proliferation of epithelium** in **apical direction**
- **Pocket epithelium** becomes **thin/ulcerated** and **bleeds more readily**
- Attempts at effective **oral hygiene become more difficult** by pocket deepening, and the cycle continues

Alveolar Bone Resorption

-Alveolar Bone Resorption: As **advancing inflammatory front** approaches the **alveolar bone**, **osteoclastic bone resorption** occurs (a **protective mechanism** to **prevent bacterial invasion** of **bone**, leading to tooth mobility/loss)

-Osteoclasts are stimulated by **proinflammatory cytokines** and other inflammatory mediators to resorb bone, and alveolar bone retreats from advancing inflammatory front

-Osteoclastic bone resorption is activated by **mediators: IL-1β, TNF-α, IL-6, PGE2**

-Receptor Activator of Nuclear Factor-κB Ligand/Osteoprotegerin: A key system for controlling bone turnover is the receptor activator of nuclear factor-κB **(RANK)/RANK ligand (RANKL)/osteoprotegerin (OPG) system**

-RANK: a **cell surface receptor** expressed by **osteoclasts**

-IL-1β and TNF-α regulate expression of RANKL and OPG, and **T-cells express RANKL**, which **binds directly** to **RANK** on osteoclast surface, **resulting in cell activation** and **differentiation to form mature osteoclasts**

-As periodontal disease progresses, collagen fibers and connective tissue attachments to tooth are destroyed, epithelial cells proliferate apically along root surface. Changes reflected clinically as **attachment loss** and/or **increased probing pocket depth**

-Changes represent an **increased inflammation severity** and **apical displacement** of inflammatory infiltrate (close to alveolar bone)

-**RANKL plays a pivotal role in bone resorption.** Involved in **osteoclast differentiation, activation**, **survival**

-Osteoprotegerin (OPG): **endogenous inhibitor of RANKL**, functions **as its decoy receptor**, **preventing RANKL binding** to specific membrane-bound receptor expressed in osteoclast precursor cells (**RANK**)

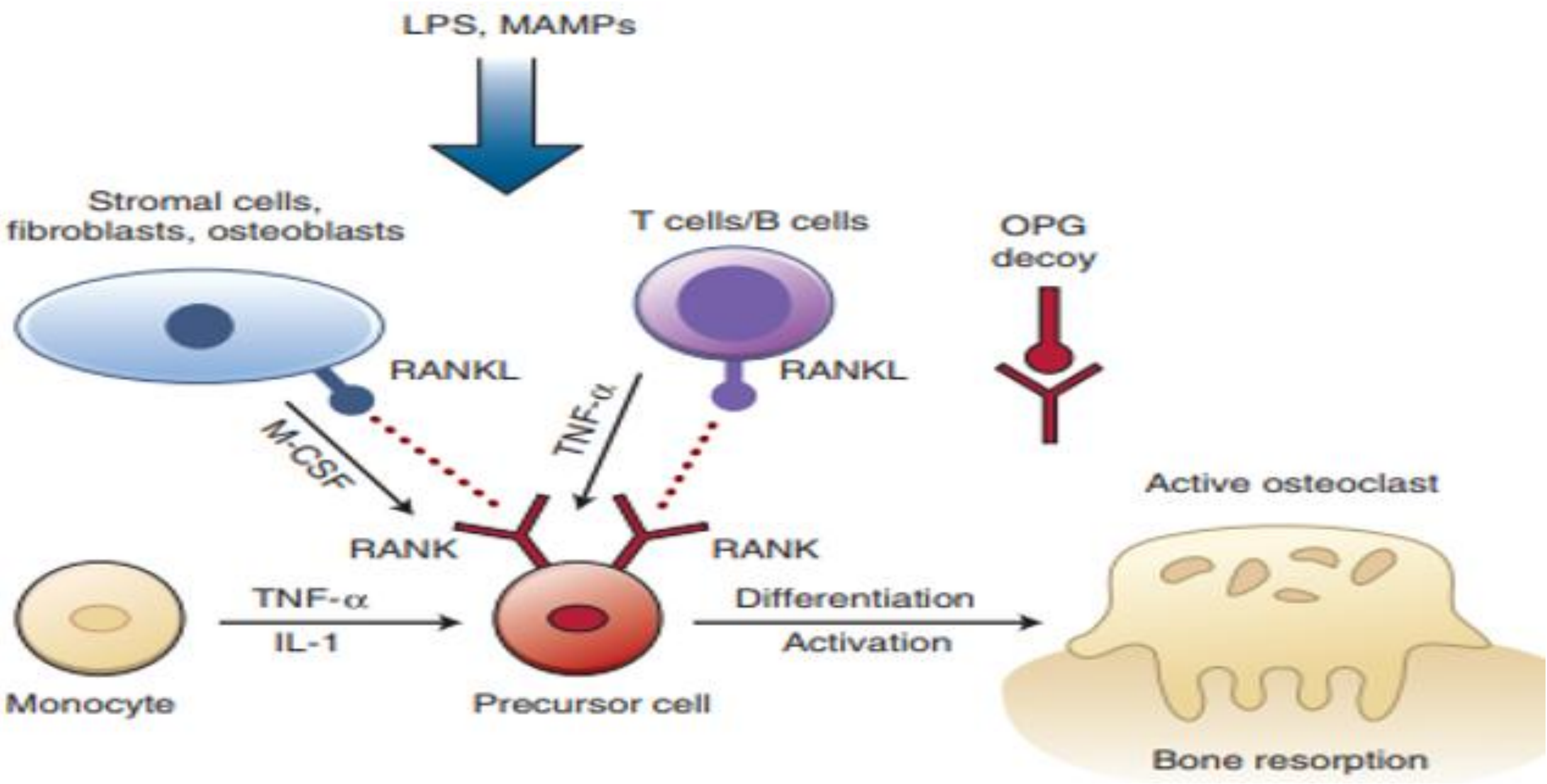
-**RANKL**: produced by **fibroblasts, osteoblasts, chondrocytes, mesenchymal cells, T- and B-lymphocytes**, whereas **OPG** is secreted primarily by **osteoblastic cells, bone marrow stromal cells, fibroblasts**

-Patients with **advanced periodontitis** have **higher levels** of **RANKL** and **lower levels** of **OPG** than periodontally **healthy** patients

-**Individual differences in host response to MAMPs and to host-derived cytokines that are the result of genetic variations, may play an important role in modulating pathogenesis of periodontal diseases.**

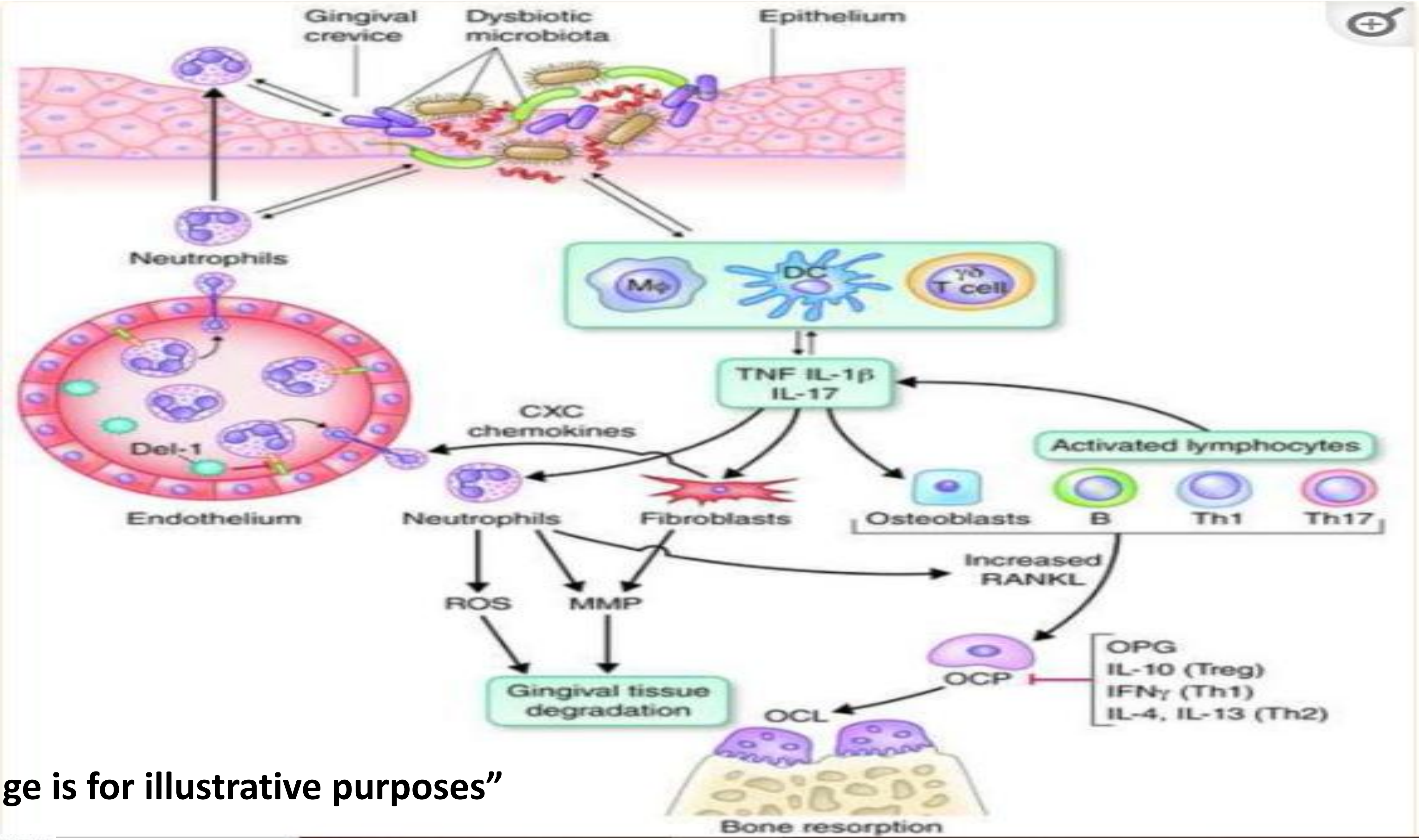
-There is a "personalized **genetic** program" controlling how each individual responds to microbial challenge, further modulated by **transitory aspects of life** that can affect production of cytokines and inflammatory mediators (**hormonal imbalances, stress, acquired diseases, medications, smoking**) which determine individual's response to microbial challenge and periodontal tissue destruction

Alveolar Bone Resorption



-**Factors regulating osteoclastogenesis: Cytokines** (e.g. IL-1, TNF-α) produced by resident and immune cells are **key** regulators. Osteoprotegerin (**OPG**) acts as a **decoy receptor** that **prevents receptor activator of (NF)-κB ligand (RANKL) from binding to its receptor activator of NF-κB (RANK)** on precursor cells and **downregulate osteoclastogenesis**

-Bacterial LPS and other **MAMPs** may **initiate** expression of **RANKL** and **osteoclastogenesis**



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Immune Responses in Periodontal Pathogenesis

- Immune response to pathogenic microorganisms is categorized as part of **innate** or **adaptive immune** system.
- **Innate** and **adaptive immunity** are **integrated** and **mutually dependent**
- **Innate Immunity:** mechanical, chemical, and microbiologic barriers preventing pathogens from invading body's cells/tissues. Saliva, GCF, epithelial keratinocytes of oral mucosa protect underlying tissues of oral cavity and periodontium
- **Commensal microflora:** may provide protection against infection by pathogenic microorganisms through competition for resources/ecologic niches and by stimulating protective immune responses
- If **primary defenses** are **breached,** then **cellular** and **molecular elements** of **innate immune response** are **activated**
- **Innate** immunity: immune response determined by **inherited factors**, **limited specificity**, **fixed**, do **not change** or **improve** during a response or with previous exposure
- **Recognition** of **pathogenic microorganisms** and **recruitment** of **effector cells** (e.g., **neutrophils**) and molecules (e.g., complement system) are central to **innate** immunity
- Recognition of periodontal pathogens and events leading to neutrophils activation in periodontium are **orchestrated** by **cytokines, chemokines**, **cell surface receptors**
- **Stimulation** of **innate** immunity leads to **inflammation**

Innate Immunity in Periodontal Diseases

- **If innate** immunity **fails to eliminate infection**, then effector cells of **adaptive immune** response (**lymphocytes**) are activated
- A challenge of **innate** immune system: **discriminate periodontal pathogens from host**, with a **limited number of cell surface receptors**. Pathogens can **mutate** to escape host recognition
- Innate immune system **recognizes evolutionarily conserved structures on pathogens** (**not** present in higher eukaryotes: PRRs)
- **MAMPs** shared by **microbes**, **not** expressed by **host** and **not** subject to **high mutation** rates
- **Pathogen Recognition** and **Activation of Cellular Innate Responses:** plaque bacteria and their products penetrate periodontal tissues, then specialized "**sentinel cells**" of immune system recognize them and signal protective immune responses. **Macrophages** and **dendritic cells** express a range of **pattern recognition receptors (PRRs)** that interact with specific molecular structures on microorganisms (**MAMPs)** to signal immune responses. **Innate** immune responses are activated providing **immediate protection**, and **adaptive** immunity is activated for a **sustained antigen-specific defense**
- **Excessive** and **inappropriate** immune responses lead to **chronic inflammation** and **tissue destruction** associated with periodontal disease

-MAMPs from microorganisms in dental biofilm activate inflammatory responses in periodontal tissues.

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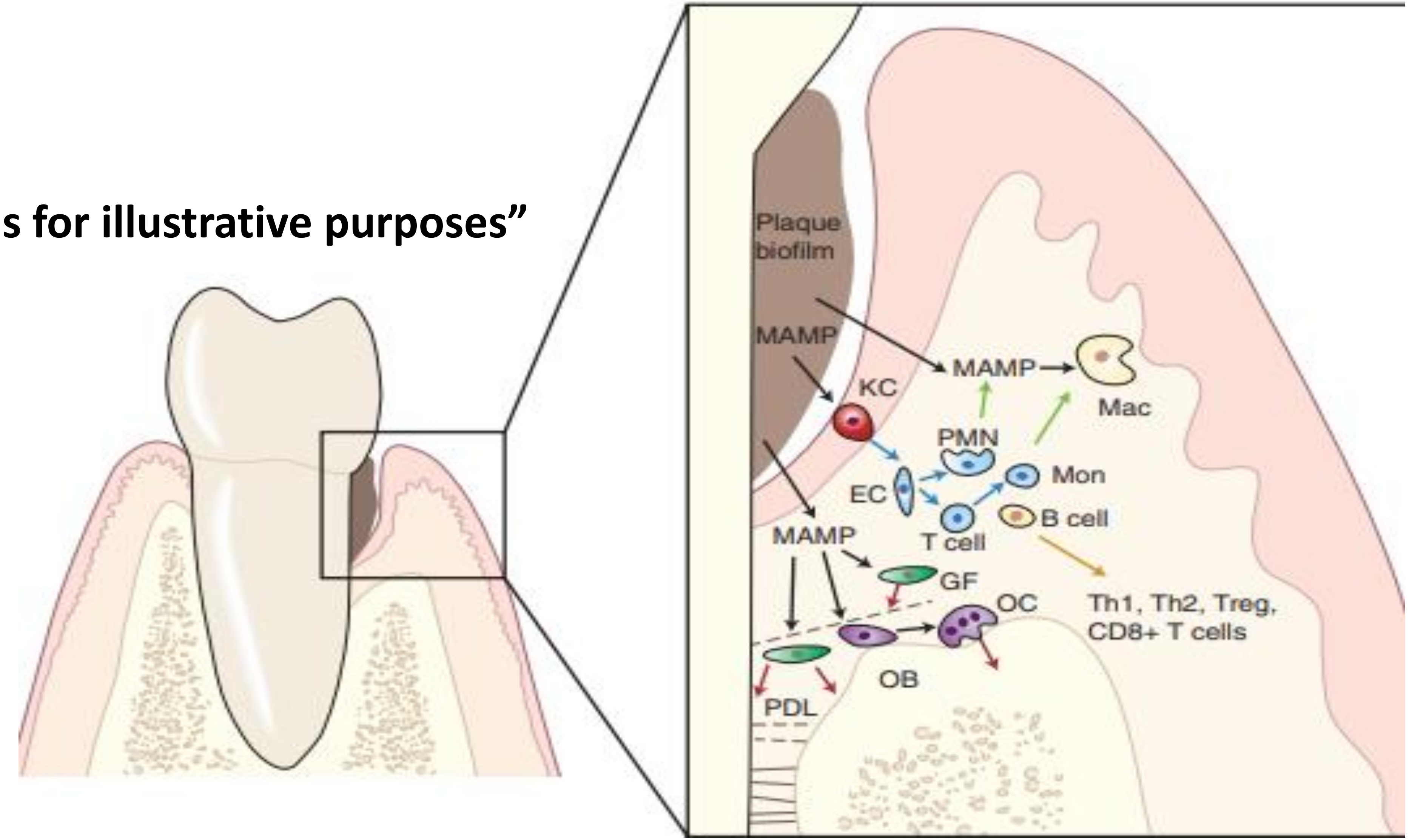


Immunity

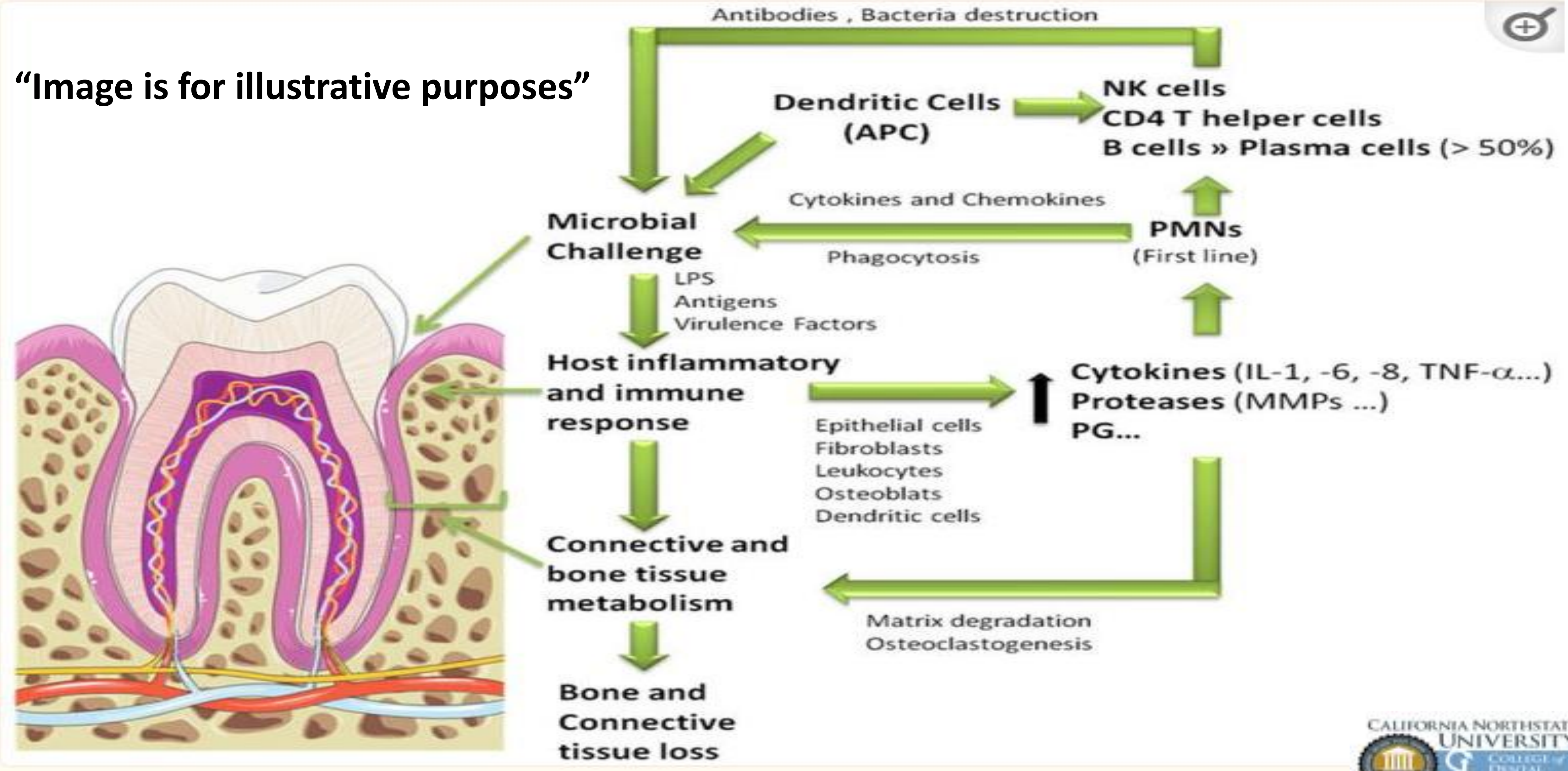
Innate Immunity in Periodontal Diseases

- **Immediate detection** of microbes and toxic products
- Triggers **inflammation** and early microbial clearance
- **Initiates** and **shapes adaptive immunity** via cytokines and antigen presentation
- Provides **rapid, nonspecific defense** during early pathogenesis

Adaptive Immunity in Periodontal Disease

- **Focused defense** against infections overwhelming innate immunity
- **Slower, coordinated activation** requiring APC–T-cell–B-cell interactions
- **Antigen-specific targeting** for precise pathogen elimination
- **Improves with repeated exposure** (memory responses)
- **Innate → adaptive shift** as disease becomes established
- **B-cells** and **plasma cells** dominate in active periodontitis, linked to pocket formation and progression

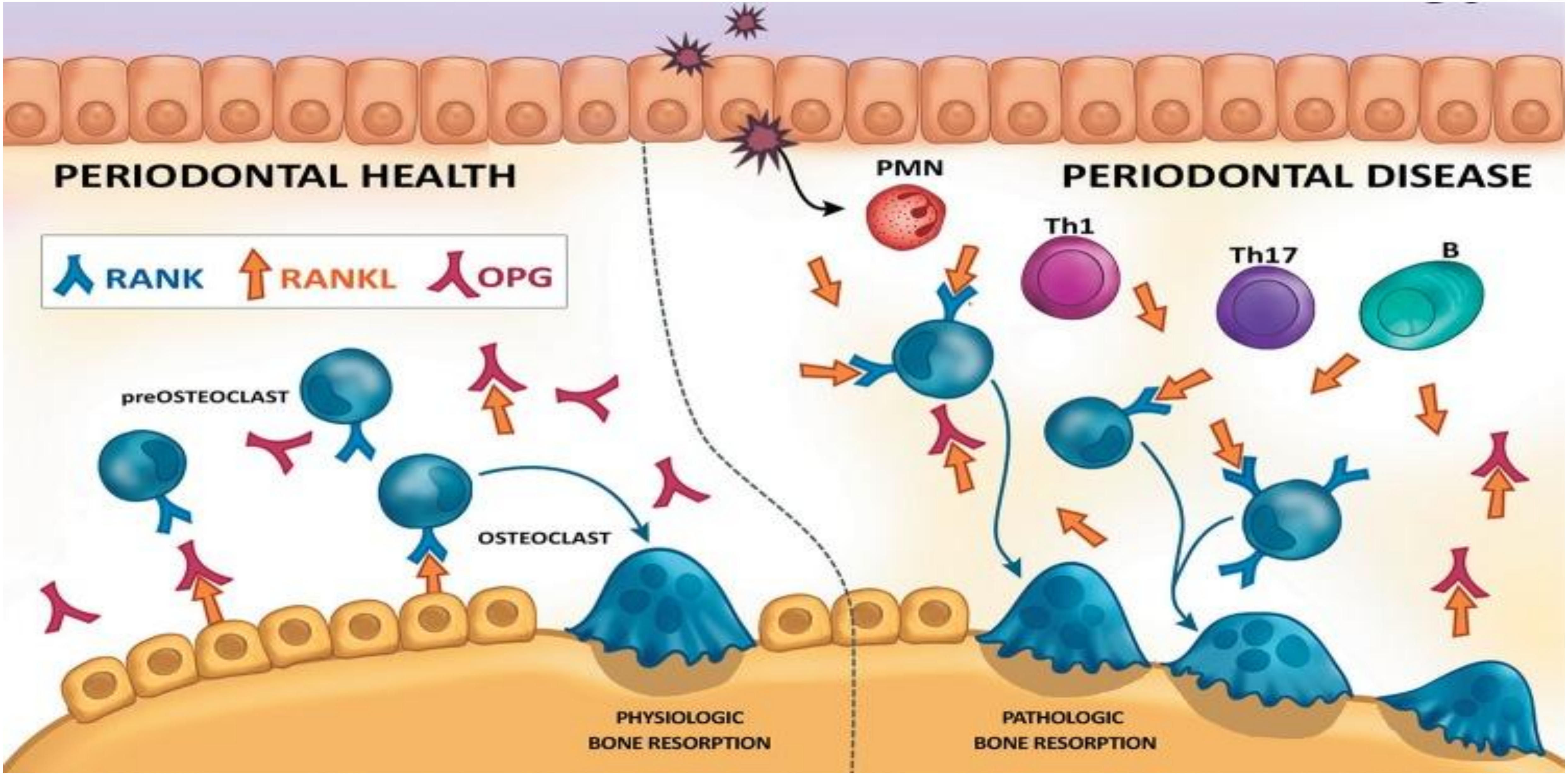
Adaptive Immunity in Periodontal Diseases



Antigen-Presenting Cells

-**Antigen presentation** drives activation of T-cells and B-cells, requiring both **APCs** and **cytokine** signals that guide T-cell development.

-The periodontium's **APCs** (macrophages, dendritic cells, B-cells) express **MHC II**, capture periodontal pathogens, and transport antigens to lymph nodes to activate effector T-cells and generate targeted immune responses.



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Antibodies

- **Specific antibodies:** produced in response to bacteria in periodontal disease, are the **endpoint** of **B-cell activation**
- **Differentiated plasma cells:** creating antibodies against bacterial antigens
- **High levels** of **antibodies** appear in **GCF** (in addition to those in circulation), and are **produced locally** by **plasma cells** in periodontal tissues
- **Antibodies** to **periodontal pathogens: IgG,** with **few IgM** or **IgA** types
- Antibodies' significance in periodontitis: **not clear**

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